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Damon Vision-Tactile Sensor Product Manual

ÿDM-Tac Wÿ

Manual Version: v3.0

Date of Completion: September 9, 2025

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Table

1. Product Overview

1.1 Product Introduction

Tactile perception is a key piece of the puzzle for realizing intelligent robots. Its addition will transform robots from simple task performers into true intelligent assistants, enabling them to make autonomous decisions and operate in complex and ever-changing environments. Compared to traditional tactile sensors, visual-tactile sensors combine the advantages of multimodal perception (including tangential force) and high resolution, aiding robots in performing delicate manipulation tasks. The DM-Tac visual-tactile sensor series from Diameng integrates high-resolution tactile perception, multimodal tactile signal measurement, and intelligent algorithms, providing users with easily accessible, high-performance tactile perception on various computing platforms.

The DM-Tac W is compatible with end effectors such as grippers, and features a rich range of sensing modes, making it a suitable end effector for grippers and other actuators. It endows humanoid tactile abilities, enabling precise control and adaptive interaction in complex scenarios.

1.2 Product Functional Features

1.2.1 Rich perceptual modalities:

It has integrated modes such as contact 3D topography, topography edge, and 3D deformation field, and supports extended measurement modes such as multi-axis force and torque, contact state, slip detection, and softness/hardness.

1. Three-dimensional morphology and morphological edges

The three-dimensional morphology refers to the three-dimensional geometric shape perceived by the DM-Tac W visual-tactile sensor on the sensing surface (as shown in Figure 1-1). When the DM-Tac W visual-tactile sensor is working, the sensor's elastic sensing surface comes into contact with the object. The sensing surface deforms due to the contact force, and the sensor can perceive the three-dimensional morphology of the contact object on the contact surface. Its three-dimensional morphology map can be displayed through a visualization window. At the same time, edge detection is performed based on the morphology map, and the edges of the contact morphology can be accurately extracted.

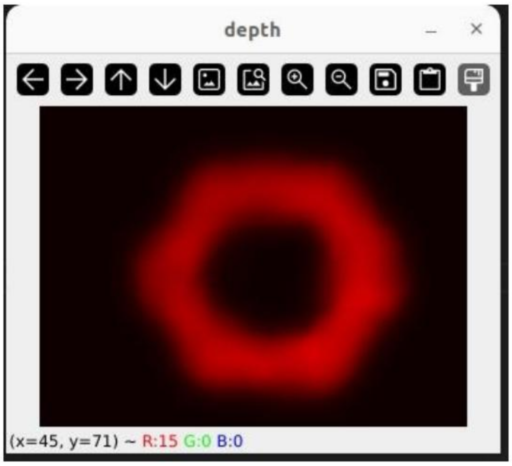

Figure 1-1 Three-dimensional topography (visualization window)

2. Three-dimensional deformation field

The three-dimensional deformation field refers to the vector field formed by the three-dimensional displacement vectors of each sensing point on the sensing surface when the DM-Tac visual-tactile sensor is working. Users can obtain the tangential deformation field and normal deformation field of the sensing surface, and the three-dimensional deformation field diagram can be displayed in the visualization window (as shown in Figure 1-2).

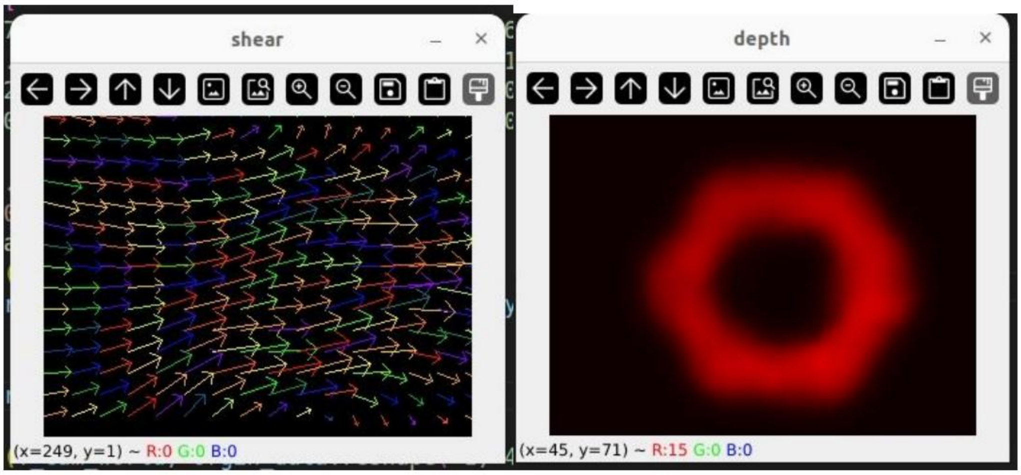

Figure 1-2 Three-dimensional deformation field (visualization window)

1.2.2 High resolution

The DM-Tac W boasts ultra-high resolution, with tens of thousands of sensing units per square centimeter. This ultra-high resolution enables...

The sensor has a strong ability to perceive the unique features of an object's shape and can capture a wealth of effective features.

1.2.3 Facilitating Precise Force Control

The DM-Tac W visual-tactile sensor boasts a 120Hz response time, providing instant force sensing.

1.2.4 Multiple models are available.

The DM-Tac W visual-tactile sensor is available in large, medium, and small sizes to suit various applications.

Products such as grippers of various sizes. A wide range of tactile properties can be achieved for different objects, materials, hardnesses, and environments.

Sensory information.

1.2.5 Signal Stability

The DM-Tac W has strong anti-electromagnetic interference capabilities, is unaffected by changes in temperature and humidity, and exhibits stable performance.

1.2.6 Easy to use

This product is easy to use and requires no calibration when replacing the sensor body or internal components.

1.3 Product Specifications

Table 1-1 DM-Tac W Product Specifications

Product Model	big	middle	Small
Size (mm) (Height × Width × Thickness)	65.4×39.0×28.2	57.4×35.0×25.0	34.3×29.0×20.1
weight	83g	61g	52g
Sensing area (mm×mm)	24×18	20×15	16×12
Sampling frequency	120Hz	120Hz	120Hz
Operating power consumption	0.71W	0.65W	0.58W
standby power consumption	0.34W	0.27W	0.23W
Camera resolution	640×480	640×480	640×480
Perceived resolution	320 × 240	320 × 240	320 × 240
Communication interface	USB2.0	USB2.0	USB2.0

Note: All specifications and parameters of this product are based on laboratory test values. Slight differences may exist between different individuals. Please refer to the official specifications for details.

Actual measurements shall prevail.

1.4 Specifications for External Dimensions, Model, and Coordinates

1.4.1 External Dimensions

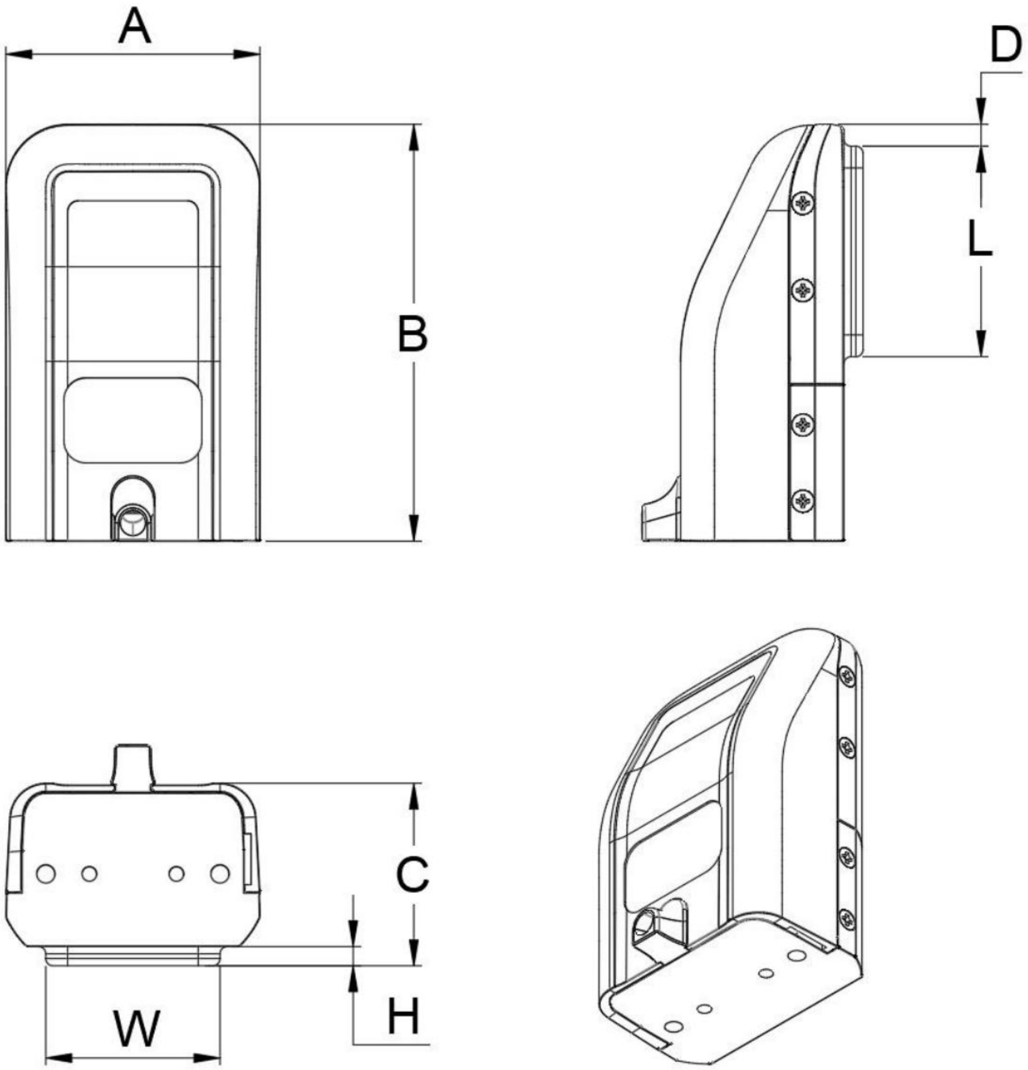

Figure 1-3 Schematic diagram of the external dimensions of DM-Tac WL and DM Tac WM

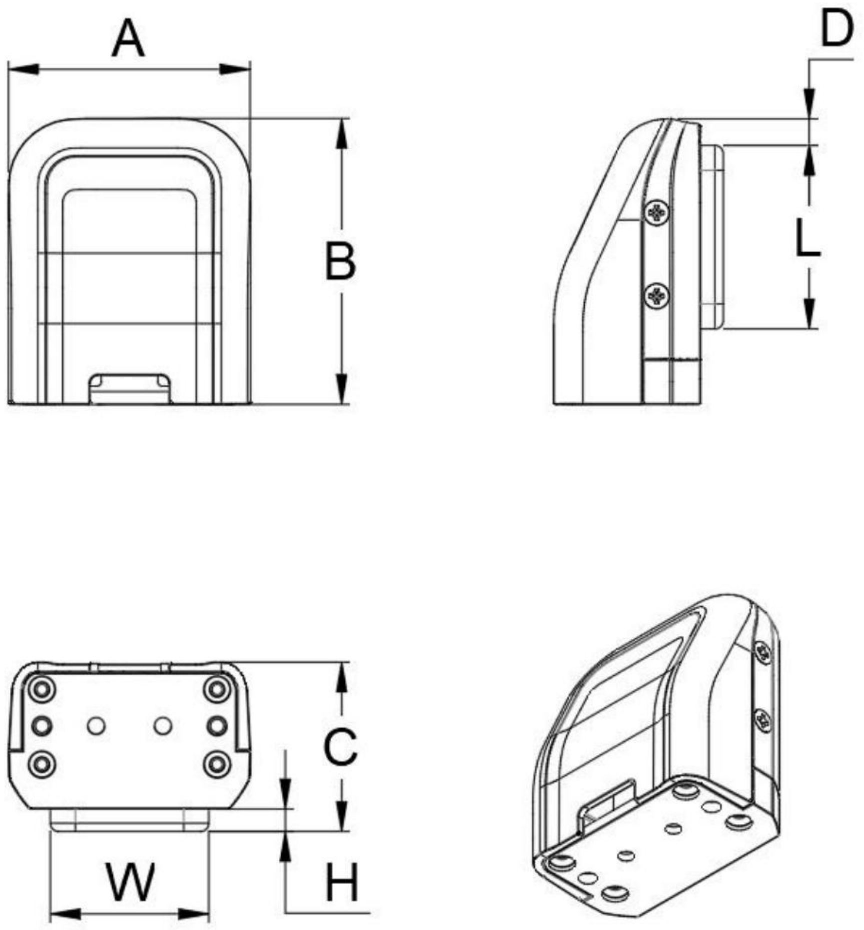

Figure 1-4 Schematic diagram of the external dimensions of DM-Tac WS

Table 1-2 DM-Tac W External Dimensions

Unit: mm

	A	B	C	D	L	W	H	drawing number
DM-Tac WL	39.00	65.35	28.15	4.25	36.00	29.00	2.65	Figure 1-3
DM-Tac WM	35.00	57.40	25.00	3.00	29.00	24.00	2.50	Figure 1-3
DM-Tac WS	28.95	34.30	20.10	3.20	22.00	19.00	2.50	Figure 1-4

1.4.2 Coordinate Specification

Figure 1-5 Schematic diagram of coordinate system specification

1.5 Hardware and software requirements

1.5.1 Software and System Environment Requirements

1. **Operating System:**
   - Windows 10 or later
   - Linux x86
2. **Python version:**
   - Python 3.8 to 3.10 (excluding 3.11 and above)
3. **Dependency libraries:**
   - NumPy version == 1.24.4
   - OpenCV Python version: 4.10.0.84
   - opencv-contrib-python version == 4.10.0.84
   - SciPy version == 1.10.1
   - setuptools version == 45.2.0
   - cupy-cuda12x version == 12.3.0 (CUDA 12.x support required)

pyudev >= 0.24.3 (Linux systems only)

1.5.2 Hardware Requirements

1. **Processor:**

Supports x86_64 architecture processors (multi-core processors are recommended for improved performance).

2. **Memory:**

Minimum requirement: 8GB RAM

3. **Storage:**

At least 1GB of free disk space is required to install the software and its dependencies.

4. **Graphics card:**

ÿ NVIDIA graphics cards that support CUDA (require the installation of the corresponding CUDA driver to support the cupy library; CUDA 12 is the default).

5. **Other:**

A network connection is required to download dependencies and updates.

ÿ Requires a USB 2.0 interface to connect the sensor.

1.6 Certification Status

It has passed FCC, CE and RoHS certifications.

1.7 Disclaimer

Usage Risks: Users should strictly comply with all applicable instructions for use and safety precautions when using this product. Use constitutes acceptance of all contents of this disclaimer. The company shall not be liable for any damage to equipment, personal injury, or third-party losses caused by the following circumstances:

• Improper use that violates the product's design purpose or operating specifications;

• Unauthorized modification of product structure or use of non-original parts;

• Equipment failures caused by failure to fulfill basic maintenance obligations;

• Unexpected damage caused by force majeure events.

Warranty Terms: The warranty and service terms for this product apply to equipment failures under normal use and require the provision of...

Complete proof of purchase. The following situations are not covered under warranty:

• Product serial number may be damaged or missing due to human error;

• Repaired by an unauthorized third-party repair shop;

• Exposure to extreme temperatures and humidity, corrosive environments, and other non-design conditions;

• Hidden damage caused by improper transportation or storage;

• Others believe it is damaged.

Product Performance: Our company is committed to ensuring the accuracy and stability of this product; all performance parameters are obtained through laboratory testing.

Tested under test conditions, we are not responsible for any functional or performance deficiencies that may occur in specific applications.

Users are responsible for assessing whether this product meets their application needs.

Intellectual Property: The design, technology, and all related materials and rights of this product belong to our company. Any unauthorized use is prohibited.

Reverse engineering, technology decryption, and unauthorized commercial use of the formula.

Limitation of Liability: Unless otherwise provided by law, the Company shall not be liable for any indirect losses, special damages or incidental damages.

Liability, including but not limited to loss of profits, loss of data, or business interruption.

The final interpretation of this disclaimer rests with Daimeng (Shenzhen) Robotics Technology Co., Ltd., to the extent permitted by law.

all.

2. Product Instructions

2.1 Packing List

Table 2-1 DM-Tac W Packing List

product	quantity	Remark
Visual-tactile sensor 1		
USB flash drive	1	The USB drive contains all the necessary files for the DM-Tac W vision-touch sensor. The document contains: • User Manual: DM-Tac W-Product Manual V3.0 20250825 • SDKÿDaimon-Tactile-Publish 20250909 • DM-Tac W-Shell Digital Model-20250417

2.2 Sensor Installation Instructions

2.2.1 Hardware Connection

The DM-Tac W series sensor connects to a computer via a USB interface. Note: The sensor requires approximately [time period missing] hours to operate. The USB transfer bandwidth is 4.5 MB/s. Due to bandwidth limitations, only one haptic sensor can be connected to a single USB 2.0 bus, and it does not support connecting multiple haptic sensors simultaneously via a USB hub.

2.2.2 Mechanical Connection

1. Connect the sensor and the robotic arm using the matching connectors.

The connector has two pre-drilled pin holes and two pre-drilled screw holes for connection to the moving parts of the robotic arm. Due to the mechanical... There is currently no universal standard for the mechanical interface reserved on one end, which may not match the opening position of the connector. During installation, the corresponding adapter can be made by 3D printing or machining.

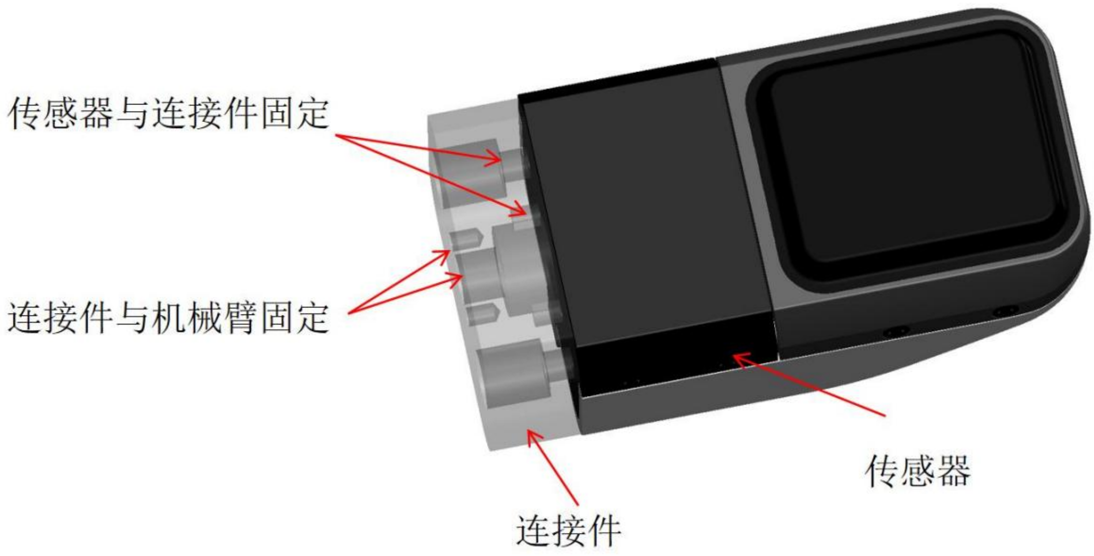

Figure 2-1 Installation position of DM-Tac W connector

2. Customized connectors

If you want to use as few connecting parts as possible when installing tactile sensors, or improve the structure of the robotic arm For compactness, you can also choose to design and manufacture your own customized connectors compatible with the tactile sensor interface. The corresponding dimensional parameters are shown in the drawings for design reference.

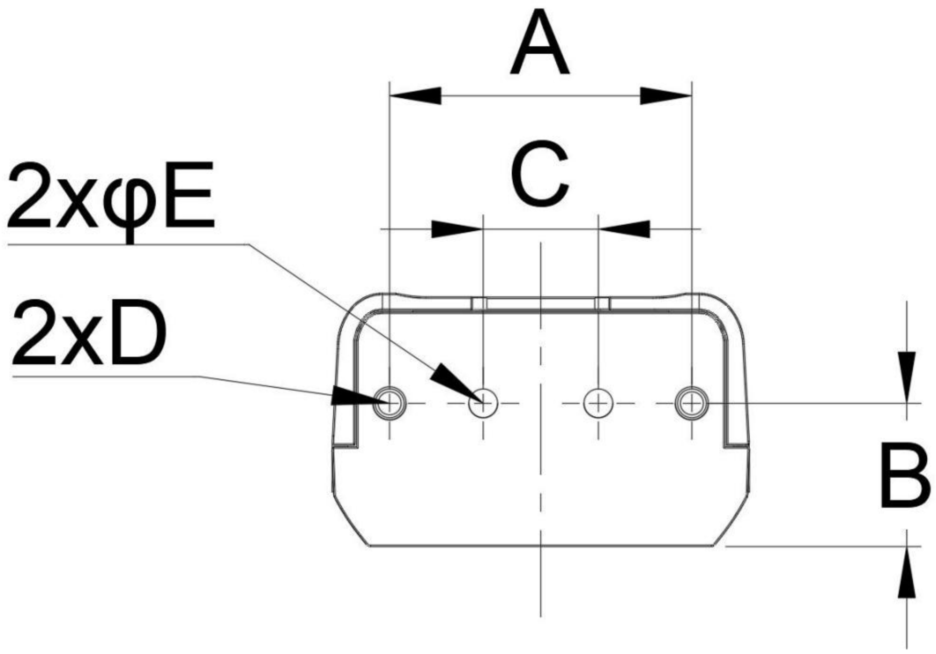

Figure 2-2 DM-Tac W connector mounting interface

Table 2-2 Connection Interface Dimensions

	A	B	C	D	ÿE
DM-Tac WL	24	11.3	8	M3x5	ÿ2 3
DM-Tac WM	24	10	12	M3x5	ÿ2 3
DM-Tac WS	21	9.85	8	M2x3	ÿ2 1

The following points should be noted when designing customized connectors:

ÿ The structure (screw holes, pin holes) of the customized connectors should be consistent with the provided parameters to ensure customization. The connectors can accurately fit with the sensors;

ÿ It is strongly recommended to use metal materials such as aluminum and stainless steel to process customized connectors to ensure the connection between the sensor and the robotic arm. The connection between them has sufficient rigidity. This is because the cantilever portion of the connector needs to withstand significant loads when using sensors. Force and bending moment. Connectors 3D printed using resin or plastic materials may exhibit significant bending during use. Curved shape.

2.2.3 Electrical Connections

The DM-Tac WS sensor's circuit board is externally mounted and connected via an FPC. Due to the characteristics of the FPC, the DM-...

The following issues should be noted when using the Tac WS sensor:

1. Avoid excessive bending, folding and stretching of FPC. The minimum bending radius should be ÿ1cm, and sudden sharp angle bends should be avoided.

2. The lifting weight of the FPC should be ÿ20g. It is recommended to fix the FPC to the moving parts.

3. The FPC should be kept away from heat sources to avoid affecting signal transmission.

2.3 Software User Manual

2.3.1 System Installation

1. Install the Ubuntu/Windows system on your computer.

2. After installation, open the terminal and use the following command to install the required dependencies:

```
pip install .
```

Tip: If the download speed is too slow, you can consider using a domestic mirror for installation.

2.3.2 Software User Manual

1. Please navigate to your local directory and open the main.py file to modify the camera serial number. You can specify the camera serial number. (String) or camera index (integer). Note: The camera serial number is an alphanumeric string containing a sequence of letters and ten digits, and it is unique to each sensor. It can be found on the yellow label on the camera's USB cable (see Figure 2-3). The camera index corresponds to /dev/videoX on Linux and follows the DirectShow (DSHOW) sorting on Windows.

Figure 2-3 Camera Identification Number

2. Next, run the main program using the following command:

```
python main.py
```

3. If the program starts successfully, it will output the data frame rate, for example:

```
Raw Measurement FPS is: 121.25
```

Typically, this frame rate should be around 120Hz, depending on your computer's performance. Four image windows will then pop up, displaying:

Original image

Overall Deformation Diagram

Normal deformation diagram (Depth)

Tangential deformation diagram (Shear)

During program execution, press 'q' to exit the program and 'r' to reset the device. Note: The device is reset by default when the sensor starts up; if the sensor fails to return to zero without contact after prolonged use, it needs to be reset. During reset, ensure the sensor surface is not in contact; do not **reset** while in contact with the sensor's contact surface .

Additionally, this code snippet will also output Output FPS. Raw Measurement FPS refers to the FPS read from the sensor.

The overall deformation information, Output FPS refers to the normal and tangential deformation information output after solving the sensor information using CUDA. If getDepth() and getShear() are not called, they will not occupy the GPU.

In this program, to reduce GPU utilization, a longer wait time can be set to slow down the data processing rate. The default delay in the script is 3ms.

```Python
k=cv2.waitKey(3)
```

2.3.3 Docker Usage Instructions

This section describes the Docker startup setup required to use sensors in Docker.

```Bash
`docker run --name xxx` (name the container) `--
    entrypoint bash -it` `--gpus all` `-v /
        Path/To/SDK:/
        Path/To/Docker` (mount the SDK code) `-v /usr/local/cuda-xxx` (mount CUDA):
        `/usr/local/cuda-xxx` `-v /dev:/dev:rw` `-v /run/udev:/run/udev:ro` `-v /sys/class/video4linux:/
        sys/class/video4linux:ro` `-
        v /sys/bus/usb:/sys/bus/usb:ro` `--device=/
        dev/video0` (set up the hardware device) `--device=/dev/video1`




        XXX (docker image name) -c "$
                {command}"
```

For visualization rendering, please first execute the following on the host machine:

```Plain Text
xhost +local:root
```

Add the following to the Docker startup command:

```Plain Text
        -e DISPLAY=$DISPLAY \ -v /
        tmp/.X11-unix:/tmp/.X11-unix \ -v $HOME/.Xauthority:/
        root/.Xauthority \
```

Use cases: Do not use this command directly; modify it according to the instructions above:

Bash

```
docker run --name DM_Tac\ --
    entrypoint bash -it \ --gpus all\ -v /
        home/daimon:/
        home/yzx \ -v /dev:/dev:rw \ -v /usr/local/
        cuda-12.6:/usr/local/
        cuda-12.6 \ -v /run/udev:/run/udev:ro \ -v /sys/class/video4linux:/sys/
        class/video4linux:ro \ -v /sys/bus/usb:/
        sys/bus/usb:ro \ -e DISPLAY=$DISPLAY \ -v /tmp/.X11-unix:/tmp/.X11-unix \ -v
        $HOME/.Xauthority:/root/.Xauthority \ --device=/
        dev/video0 \ --device=/dev/
        video1 \ ros1-cuda:latest \ -c "${command}"
```

2.4 SDK User Guide

This section details the key functions in this software. The function's purpose, usage, and return value are listed to help users fully understand and effectively utilize these features.

2.4.1 Sensor()

• Function: Instantiates the sensor. Sets `dev_serial_id` to either the camera serial number (string) or the camera index (integer). On Linux, the camera index corresponds to `/dev/videoX`; on Windows, the index follows DirectShow sorting. The serial number is an alphanumeric string, consisting of a sequence of letters and ten digits, printed on a yellow label on the camera's USB cable (see Figure 2-3). By default, `KEEP_FPS_Print` is...
False; setting it to True will cause the program to print the camera's frame rate every two seconds.

• **How to use:**

Python

```python
sensor = Sensor(dev_serial_id, KEEP_FPS_Print=False)
```

• **Return value:** This function has no return value.

2.4.2 reset()

• **Function:** Resets the hardware, clears the current state, and restores the device to its initial state. Ensure the sensor surface is not in contact when performing the reset operation.

• How to use:

```python
sensor.reset()
```

• **Return value:** This function has no return value.

2.4.3 getRawImage()

• **Function:** Acquires raw image data captured by hardware devices, suitable for image processing or visual analysis applications. Please note that this image typically lacks practical application value, and differences may exist between different sensors; it cannot be directly used as input for end-to-end model training in neural networks. It is recommended to use processed 2D or 3D deformation information as measurement data. However, the raw image can be used by users to assess and adjust sensor status to identify faults such as wear and tear or light failure; therefore, this function is provided.

• How to use:

```python
image = sensor.getRawImage()
```

• **Return value:** Returns an array of shape (240, 320) with data type uint8.

2.4.4 getDeformation2D()

• **Function:** Extracts two-dimensional overall deformation data from the sensor from the original image for analyzing surface deformation.

• How to use:

```python
deformation = sensor.getDeformation2D()
```

• **Return value:** Returns an array of shape (240, 320, 2) with data type float32, representing the deformation information of each pixel in millimeters (mm).

2.4.5 getDepth()

• **Function:** Acquires normal deformation data detected by the sensor. Normal deformation refers to elastic deformation perpendicular to the contact surface. (This function calls CUDA for calculation).

• **How to use:**

```python
depth = sensor.getDepth()
```

• **Return value:** Returns an array of shape (240, 320) with data type float32, representing the normal deformation information of each pixel in millimeters (mm).

2.4.6 getShear()

• **Function:** Acquires tangential deformation data detected by the sensor. Tangential deformation refers to deformation in a direction parallel to the contact surface . (This function calls CUDA for calculation).

• **How to use:**

```python
shear = sensor.getShear()
```

• **Return value:** Returns an array of shape (240, 320, 2) with data type float32, representing the tangential deformation information of each pixel in millimeters (mm).

2.4.7 put_arrows_on_image(image, arrows, scale = 1.0)

• **Function:** Used for visualization of two-dimensional deformation fields; the scale parameter adjusts the arrow size.

• **How to use:**

```python
cv2.imshow('deformation', put_arrows_on_image(black_img, deformation*20))


cv2.imshow('shear', put_arrows_on_image(black_img, shear*20))
```

Return value: Returns an array of shape (240, 320, 3) with data type uint8. Normally, it will display an image with arrows.

2.4.8 getCameraID()

• **Function:** Obtain the camera serial number of the sensor.

• **How to use:**

```python
```

```
shear = sensor.getCameraID()
```

Return value: Serial number (string).

2.4.9 getStatus()

• **Function:** Used to report the current device status and return one of the following SensorStatus constants: OK (0), RESETTING (1) or DISCONNECTED (2). Internally, it returns 0 if the sensor exists and is not being reset; 1 if it exists but is being reset; and 2 if it does not exist.

• **How to use:**

```
Python
shear = sensor.getStatus()
```

Return value: 0/1/2 (int).

2.4.10 disconnect()

• **Function:** Release sensor.

• **How to use:**

```
Python
shear = sensor.disconnect()
```

Return value: None.

Notice

Before calling the above functions, ensure that the hardware device is correctly connected and initialized. If any exceptions or errors are encountered during the call, please check the hardware connection and the normal working status of the drivers. To ensure optimal performance and compatibility, it is recommended to use fixed versions of dependency libraries, avoiding frequent updates.

3. Safety Precautions

3.1 Hardware Usage Precautions

1. The DM-Tac WS sensor's circuit board is externally mounted and connected via an FPC. Due to the characteristics of the FPC, the following issues should be noted when using the DM-Tac WS sensor:

Avoid excessive bending, folding and stretching of FPC. The minimum bending radius should be ÿ1cm, and sudden sharp angle bends should be avoided.

The load capacity of the FPC should be ÿ20g, and it is recommended to fix the FPC to the moving parts.

The FPC should be kept away from heat sources to avoid affecting signal transmission.

2. Force Application Range: When

operating the sensor, please ensure that the force applied to the sensing surface is always within the safe range specified in this manual to avoid exceeding the specified value and causing equipment damage. Sensor load-bearing capacity range: Resultant force not exceeding **20.0N, not less than...**

Within 0.2N (the sensor cannot detect anything less than 0.2N).

2. Area of force application

When operating the sensor, please apply force within the sensing area specified in this manual. If the applied force exceeds the specified sensing area, it may result in data loss, distortion, or other issues, and accurate and effective sensing information cannot be guaranteed.

3. Avoid damage to the surface from sharp objects.

The silicone surface is soft and easily scratched or cut by sharp objects. Avoid friction or collision with hard objects during use to protect the surface integrity.

4. Risks of use in high-temperature environments

Please avoid exposing the sensor to temperatures exceeding **50°C .** Excessive heat may impair sensor performance.

Decreasing temperature, material aging, or even malfunction can occur. Before use, ensure the sensor is operated within the specified operating temperature range.

5. Prevent electrical overload

If the sensor overheats or emits smoke during use, immediately disconnect the power and stop using it. Inspect the sensor.

Is there an electrical overload or fault?

6. Do not use organic solvents or liquids to clean the sensing surfaces.

Do not allow alcohol or other organic solvents to come into contact with the sensor. Silicone is sensitive to chemical solvents such as alcohol, acetone, and paint. Contact with the sensor may cause surface damage, leading to sensor malfunction.

7. Prevent severe physical shocks. Because the

sensor contains delicate electronic components, please avoid exposing it to severe vibrations or drops. Drops or strong impacts may cause the sensor to malfunction or be damaged.

3.2 Precautions for Software Operation

1. If multiple cameras are connected to the computer, the camera ID of the sensor may need to be checked and set by the user.

Users need to enter either the camera serial number (a string of letters and numbers, located on the yellow label on the sensor's USB cable, attached to the sensor) or the camera index (0, 1, 2, ...). If no images are acquired, please verify that the entered serial number is correct. The software will automatically detect available devices upon startup and display their serial numbers and camera indexes.

2. Reset Precautions

Reset is not possible while the sensor is in contact with its surface.

3. Cupy installation issues

When installing Cupy, you may encounter issues with CUDA version incompatibility. Please select the appropriate Cupy version based on the CUDA version installed on your system.

If you are using CUDA 12, please install cupy-cuda12x.

If you are using CUDA 11, please install cupy-cuda11x.

If you encounter errors when installing using pip, it is recommended to try installing Cupy using conda: `conda install anaconda::cupy`, to avoid dependency issues.

By using the methods above, you can resolve common camera and Cupy installation issues, ensuring the software functions correctly.

3.3 Safety Early Warning

1. Prevent children from misusing it.

This sensor contains small parts and poses a risk of accidental ingestion. Please keep the sensor out of reach of children to prevent accidental swallowing or misuse. If a part is swallowed, seek medical attention immediately.

3.4 Maintenance and Care

3.4.1 Regular inspection and cleaning

1. To ensure the long-term stability and reliability of the sensor, please regularly check its appearance and operating status. Periodically check the wiring, connection ports, and housing for damage or looseness.

2. When cleaning, use a slightly damp soft cloth or sponge to wipe the sensor surface. Avoid using abrasive cleaning tools to prevent scratching the sensor surface.

3. When cleaning, avoid using organic solvents such as alcohol and acetone directly, as these may damage silicone or plastic parts. Neutral cleaning agents are recommended.

3.4.2 Temperature and Humidity Control

1. When storing the sensor, please maintain the ambient temperature between **0°C and 50°C ,** and the humidity between **30% and 90%** relative humidity . Avoid extreme temperature and humidity environments, as these can affect the sensor's accuracy and lifespan.

It has a negative impact on life.

2. Avoid exposing the sensor to direct sunlight.

3.4.3 Prevention of pollution and dust

1. When storing the sensor for an extended period, ensure it is kept in a clean, dry environment. Prevent dust or foreign objects from entering the sensor's contact ports or sensitive areas.

2. If the sensor is not used for a long period of time, it is recommended to put it in its packaging box or use a dust cover to reduce the impact of external dust on the device.

3.4.4 Regular functional testing

1. Before each use, it is recommended to perform a brief functional test on the sensor to ensure it is working properly. During the test, the sensor's response and data output can be detected by simulating actual operation.

2. If any abnormalities are found (such as slow response, inaccurate data, etc.), please troubleshoot immediately.

3.4.5 Avoid physical damage

1. During handling, storage, or use, avoid applying excessive external force or impact to the sensor. Severe impacts may cause the sensor to malfunction.

2. When using, please ensure that the sensor is securely installed and avoid any undue pressure on the sensing surface of the sensor.

3.4.6 Replacement of the perception layer

1. The sensor's sensing layer contact surface is made of silicone. After prolonged use, wear or aging may occur, affecting its performance. To ensure the sensor's normal operation, it is recommended to regularly check the condition of the silicone layer and replace it promptly if cracks, hardening, or other damage are found.

2. No calibration is required when replacing the sensing layer. Simply unscrew the screws securing the upper layer of the sensor to remove and replace the sensing layer, **as shown in Figure 3-1. Note: When replacing the sensing layer, do not use excessively long screws to avoid damaging the internal components of the sensor. The standard screw size included with the DM-Tac W is M1.6*2.5.**

Figure 3-1 Schematic diagram of perception layer replacement

3. When replacing, ensure that there are no contaminants or dust inside the sensor, and avoid damaging other parts of the sensor during the operation.

Caused damage.

4. Contact Information

Email: info@dmrobot.com

Telephone: 0755-83087156

5 Test Parameters

The machine has a full	20N (1cm² contact area)
load range,	40N (No significant performance difference after recovery within 1 hour)
maximum	<1%FS or 0.02mm compression depth
overload	<1%FS
sensitivity, zero	<2%FS
drift, and non-linear output.	<3%FS

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